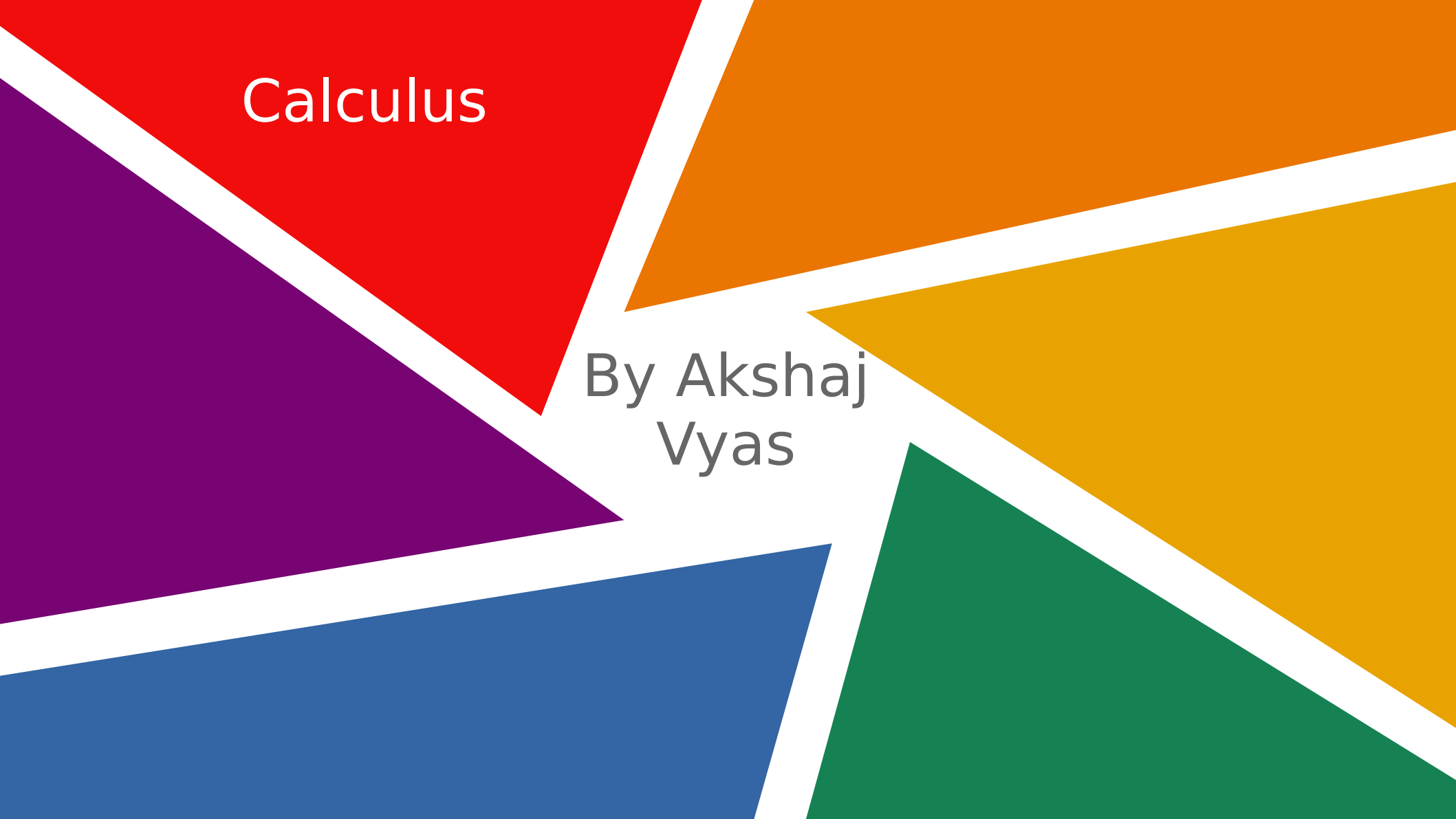


The background consists of several large, overlapping triangles in various colors: red, orange, yellow, green, blue, and purple. The triangles are arranged in a way that they seem to radiate from a central point, creating a dynamic and colorful composition. The colors are vibrant and the shapes are sharp.

Calculus

By Akshaj
Vyas

What's a Limit?

- When a function becomes arbitrarily close to a value at a point, but is undefined when evaluated at that point.
- $f(x) = (x - 2)/(x - 2)$
- $f(2.01) = 1$, but $f(2) = \text{undefined}???$
- But the graph tells us it should be 1!

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- We now call this $\lim_{x \rightarrow 2} \left(\frac{x - 2}{x - 2} \right) = 1$

Differential Calculus

- “Instantaneous” rate of change!



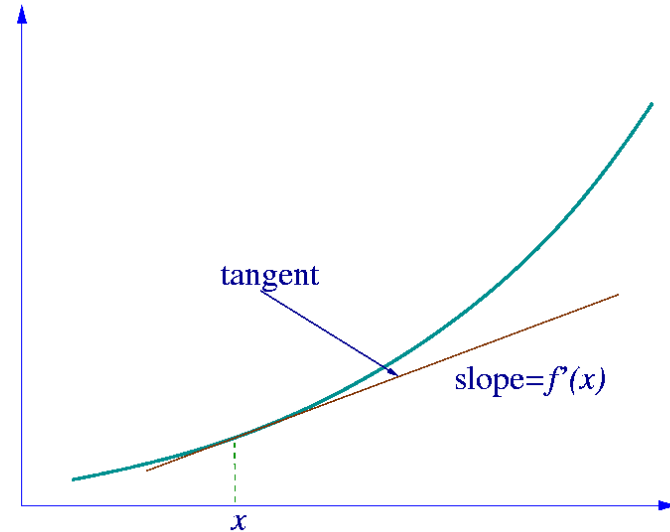
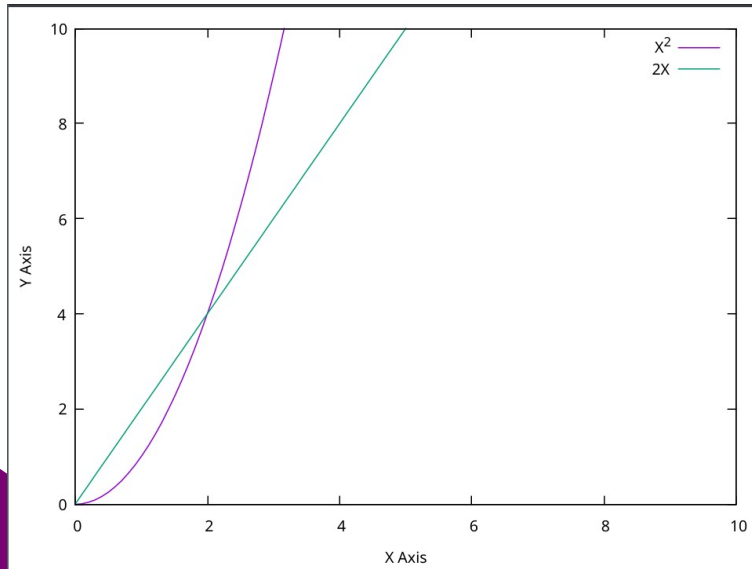
Differential Calculus

- How can we resolve this paradox?
- By using a limit!

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Differential Calculus

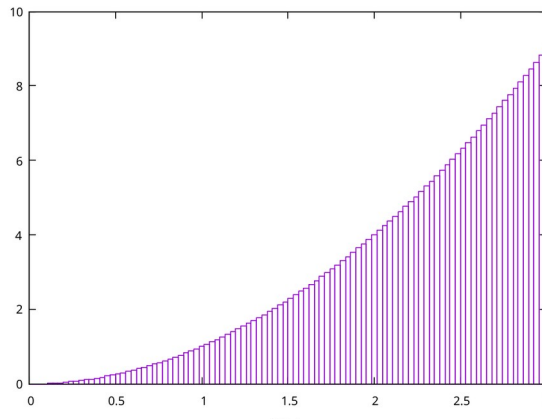
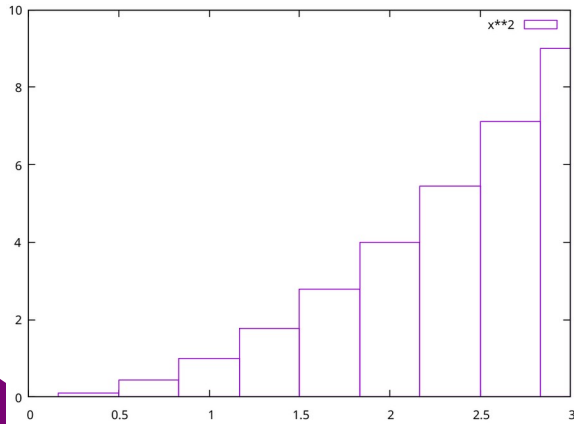
- Slope of tangent line of graph.



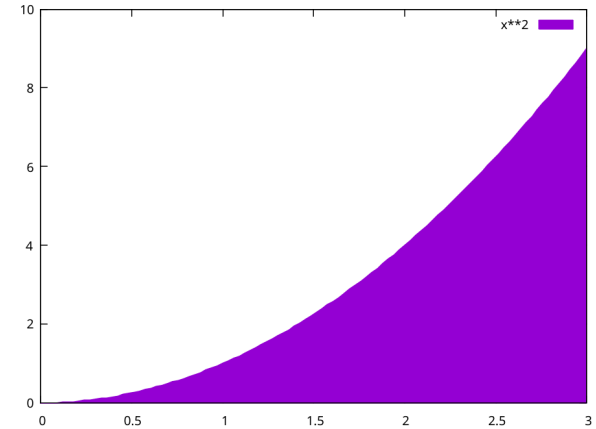
Integral Calculus

- Area under the curve

Approximate



Exact



Wait... They're opposites??

- Yes, taking the derivative of an integral yields the original function
- Second Fundamental Theorem of Calculus:

$$\frac{d}{dx} \int_a^x f(t) dt = f(x)$$